StructTutor Progress Review: 30 October 2026 Version

Of the 29 items in my first report, 13 are fixed, 8 are partly fixed and 8 are not fixed. The first launch is now honest and far tidier, and the app covers all 20 modules. The most serious problem is new: in every multiple-choice question the correct answer is the first option and almost always the longest, so the exams can be passed without knowing the material.

1. How I checked

I unzipped the project, installed it, built it, ran the 12 automated tests and the type-check, dumped all 710 questions into a list for analysis, recomputed 12 numeric answers by hand, read the grading and storage code, and studied the saved screenshots. I did not click through the live app, so anything about feel or animation is from screenshots only.

Health check results. The type-check is clean. All 12 tests pass. The build produces one 1.12 MB file (dist/StructTutor.html). A plain `npm install` fails on a clean machine (see N11), so I used a workaround to build.

2. Scorecard against my first report

Status key: **Fixed**, **Partly**, **Not fixed**.

ID	Item	Status	Evidence
A1	Developer button on every screen	Fixed	Not visible in any screenshot. The drawer component still exists in UtilityModals but nothing uses it.
A2	Fake progress on dashboard	Fixed	Starts at 0%, 0 of 20 modules, 'No sessions recorded yet', recent sessions list empty by design.
A3	XP counter	Fixed	Gone from the interface. Only a test mentions it.
A4	Emoji mixed with icons	Fixed	None in the screens. A few remain in code comments and internal labels.
A5	Internal jargon	Partly	'Reconstruction' gone from view. 'Milestone A to F Portfolio Tracker', 'Course Gate Progress' and 'Verified Activity Log' remain. They now match your syllabus, but read heavily.
A6	Tutor header four rows tall	Not fixed	Still three stacked panels before the question. The diagram only just fits at about 870 px of height and would be cut off at 768 px.
A7	Repeated Details buttons	Partly	Learning map rows now show Details, Study Topic and Topic Assessment. Roles differ, but it is still

ID	Item	Status	Evidence
			three buttons per row.
A8	False precision	Not fixed	Question 2a still shows 666,666.7 mm³.
A9	Browser tab and metadata	Fixed	Title, description and sharing tags now describe StructTutor.
A10	Light and dark theme	Partly	A switch exists, but it only toggles card colour (Hybrid Light Cards or Dark Cards). The page stays dark.
A11	Keyboard use	Not fixed	No key handlers or tab-order code anywhere. The assessment screen has no accessibility labels at all.
A12	Fonts from the internet	Not fixed	index.html still loads three font families from Google.
B1	Saving progress	Partly	Assessment results are saved (browser localStorage). Tutor practice is not saved. See N7.
B2	Two apps, one product	Fixed	Python app frozen in legacy-python and clearly labelled.
B3	Content out of code	Not fixed	Worse: about 739 KB of content now sits in five TypeScript files. No schema checking.
B4	Giant main file	Partly	Three views were extracted, but App.tsx grew from 2,108 to 2,691 lines and still has 32 pieces of state.
B5	Dead code	Fixed	Three files removed after porting the useful diagram modes.
B6	General beam engine	Not fixed	No engine. Diagrams are fixed illustrations, not drawn from calculated results.
B7	Review status and notice	Partly	Labels and a footer notice exist, but the labels are applied automatically. See N5.
B8	Unbuilt features stated	Fixed	README and known-limitations say PDF import and the AI tutor are not built.
B9	Content template	Not fixed	No template or guide in docs.
C1	One launcher	Partly	One launcher, but it calls `npm install`, which fails on a clean machine.
C2	One built file	Fixed	Single build command, one output file, version shown in the sidebar.

ID	Item	Status	Evidence
C3	Clutter	Fixed	Old zips and stray files removed. A 362 KB syllabus PDF and 14 screenshots remain, which is fine.
C4	Unused packages and boilerplate	Fixed	Four packages removed, README rewritten, Gemini boilerplate deleted.
C5	Tests	Partly	One working `npm test` with 12 tests. See N8 for their weakness.
C6	Out-of-date counts in docs	Fixed	No hard-coded counts.
C7	Known limitations recorded	Fixed	docs/KNOWN_LIMITATIONS.md lists the five Python catch-all blocks.
C8	Settings, export, import	Not fixed	Only Reset Progress exists. No export, no import, no settings page.

3. New problems I found

N1. Multiple-choice exams can be passed by clicking the first option (critical)

All 320 multiple-choice questions store the correct answer as option 1 (the first). I read the display code and found no shuffling. In 316 of 320 questions (99%) the correct option is also the longest; by chance it would be about 25%. Wrong options are often absurd, for example 'Because gravity acts horizontally on intermediate columns'.

Why it matters. The 90% pass rule and the mastery percentage mean nothing if the exam can be gamed. It also teaches test-taking tricks rather than engineering.

Fix. Shuffle options on every attempt (store the shuffle seed with the attempt). Rewrite distractors so they are plausible, similar in length and based on real misconceptions (for example a sign error or a forgotten factor of 2). Add an automatic test that fails if the correct answer is the longest option in more than about 35% of questions.

Size. Small for the shuffle, medium for the distractor rewrite.

N2. The question bank is wide but shallow and formulaic

There are 710 questions: exactly 5 per topic (62 topics) and 20 per module exam. Modules 0 to 6 have 250 in total (35, 40, 35, 35, 35, 35, 35). They are fixed questions with no variants, so a second attempt shows the same five. Only two types are used (numeric and multiple choice; a 'scenario' type exists but nothing uses it). None has hints. Explanations are short: median 129 characters, and 133 are under 60 characters. Only 25 questions mention a book, standard or clause anywhere in their text, and none has a structured citation.

Why it matters. You asked for many cases per module and a very large bank. At present each sub-skill is tested once with one fixed example.

Fix. Move to question templates with generators (each producing dozens of valid variants), case families per objective, and stored sources. This is the main purpose of the revised prompt pack.

Size. Large.

N3. Numeric answers: a good sign

I recomputed 12 randomly chosen numeric answers by hand across Modules 0, 1, 2, 4, 5 and 6 (reactions, bending moment at a point, section modulus, second moment of area by two methods, bearing pressure, Euler ratio). All 12 were correct and the explanations show working. This is a sample of 390 numeric items, so it is encouraging, not proof.

N4. How numeric answers are marked

The grader reads only the number typed in. The unit is shown beside the box but never entered or checked, so typing 12.5 for a kN answer and for an N answer is treated the same. Tolerance is 1.5% of the answer. Every attempt uses the same five questions, and a 90% pass mark on five questions means all five must be right, so repeat attempts reward memorising.

Fix. Accept typed units and convert them (the tutor engine already does this for its own questions), give targeted feedback when a unit or a factor of 1,000 is wrong, and draw each attempt from a larger bank.

Size. Medium.

N5. Review status is applied automatically, not earned

Every one of the 390 numeric questions is labelled 'Self-Checked Calculation' because the helper function that builds them sets that label by default. Nothing runs a check. All 710 questions also carry the same notice saying 'first-principles mechanics verified'. For safety-related learning material that is an unsupported claim.

Fix. Default every item to Draft. Only the build process may promote an item, and only when an independent check actually passed. Reword the notice. Add a Review Queue where you mark items Reviewed or Wrong.

Size. Medium.

N6. Teaching content is thin outside Topic 0.1

Only Topic 0.1 has the interactive tutor with hints (eight steps). The other 61 topics have a summary card, a five-question test and worked-example briefs for self-review. This is expected while you have not supplied the books, but it means the app currently tests more than it teaches.

N7. Saved progress can be silently wiped

Progress is stored under a revision ID. If the stored revision differs from the current one, the code ignores the saved data and starts fresh, and the next save overwrites it. Because you will keep revising content, one change to that ID would erase everything. There is also no export or backup. Browsers also

keep storage separately by file location when you open a local HTML file, so moving or renaming the file may look like lost progress (I have not tested this).

Fix. Never discard on a revision change. Migrate instead. Add Export and Import, and move to IndexedDB for larger data.

Size. Medium.

N8. The tests prove little about correctness

The main test checks that each question marks its own stored answer as correct. That is true by construction, so it cannot catch a wrong stored answer. No test compares answers with the Python reference, and there are no tests of the screens.

Fix. Compute answer keys from engine code and add golden-value tests against the Python reference. Add the shuffle and distractor tests from N1.

Size. Medium.

N9. Interface bug on the tutor question page

In the Question 2a screenshot the instruction 'Enter your answer in any listed stress unit...' appears inside the formula box after an equals sign, as if it were part of the equation. Earlier it sat beside the box.

Size. Small.

N10. Two imperial-unit leftovers and no enforcement

The Module 11 exam mentions a '215 mm (9-inch)' brick wall, and a Module 8 option refers to 'inches and millimetres'. Your metric rule is otherwise respected, but the README says 'throughout' and nothing checks automatically.

Fix. Remove both, and add the metric linter to the test command.

Size. Small.

N11. A clean install fails

On a machine with no saved packages, `npm install` stops with a version conflict (esbuild 0.25 pinned in devDependencies against what Vite 8 allows). run_windows.bat calls `npm install`, so a fresh machine would fail on first launch. I tested a copy with the esbuild line removed and the install then succeeded; I did not rebuild that copy.

Fix. Remove the esbuild line from package.json and commit a lock file.

Size. Small.

N12. Smaller structural risks

- The question type carries duplicate fields (type and questionType, unit and requiredUnit, tolerance and toleranceAbs, assumptions and assumptionsAndLimitations), and the two helper

functions accept two different argument orders. This invites mistakes when thousands of items are added.

- 88 of 710 prompts depend on other items ('the same beam', 'Syllabus Example 4.1'). That breaks as soon as questions are shuffled or drawn at random.
- The changelog covers only Checkpoint 1 and the version is still 0.1.0. The build date is a typed constant, so it will not update by itself.
- All content loads in one 1.05 MB script. Fine today, but content should load per module as it grows.
- The unused verification drawer should be deleted or moved into a Diagnostics page.

4. What is good (objectively)

- The dashboard is honest and calm: 0% mastery, an empty activity log with a clear next step, Reset Progress, version number and an educational notice.
- The old prototype problems are gone: no XP, no floating developer button, no emoji in the screens, a proper tab title, one launcher, one built file, no dead code or clutter.
- All 20 modules exist, with 62 topics, topic assessments, 20 module exams, a 90% pass rule with boundary tests, a coverage and milestones view, and a learning map with stages, search and filters.
- Sampled numeric answers were correct and show their working.
- The type-check is clean and the 12 tests pass.

5. Against your newer requirements

Requirement	Status
Very large question bank	Not met. 710 fixed questions, 250 for Modules 0 to 6.
Many cases per module, proficiency in everything	Not met. One fixed question per sub-skill, no variants, no case coverage tracking.
Multi-step design problems with non-unique answers (Eurocode)	Not built. Only numeric and multiple-choice types exist.
Understands learning pace	Not built. No adaptive difficulty, timing or spaced review.
Chat and RAG over the books	Not built. No library, no chat, no model connection.
Metric only	Mostly met, with two exceptions and no automatic check.
Smart marking	Not met. Number-only marking, no unit checking, same questions each attempt.

6. Recommended order of work

Do first (about one to two days, before any more content).

1. Shuffle options and rewrite weak distractors (N1).
2. Fix the clean-install failure (N11) and the two imperial items (N10).
3. Stop auto-labelling items as self-checked and reword the notice (N5).
4. Make saved progress safe: no wipe on revision change, plus Export and Import (N7, C8).
5. Fix the formula-box bug (N9) and the 666,666.7 display (A8).

Foundations before scaling content (several days).

6. Move content out of TypeScript into checked files with one question format (B3, N12).
7. Add templates, generators and case families so one item yields many variants (N2).
8. Unit-aware marking and random draws from a bigger bank (N4).
9. Engine-computed answer keys and golden tests (N8, B6).

Then the new capabilities. Pace model and planner, multi-step design tasks, then the library and chat.

Polish. Tutor header (A6), keyboard use (A11), local fonts (A12), full theme (A10), jargon (A5), changelog and version.

7. What I could not verify

I did not run the live app, so I could not confirm that the multiple-choice order stays fixed at runtime (I read the code, which shows no shuffling), how file location affects saved progress, or how the screens behave at other window sizes. I hand-checked 12 of 390 numeric answers and none of the 320 multiple-choice answers beyond reading samples. I have not judged the correctness of Modules 7 to 19 content in depth.